



Haar wavelet downsampling: A simple but effective downsampling module for semantic segmentation

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ARTICLE INFO

Keywords:

Semantic segmentation
Downsampling
Haar wavelet
Information entropy

ABSTRACT

Downsampling operations such as max pooling or strided convolution are ubiquitously utilized in Convolutional Neural Networks (CNNs) to aggregate local features, enlarge receptive field, and minimize computational overhead. However, for a semantic segmentation task, pooling features over the local neighbourhood may result in the loss of important spatial information, which is conducive for pixel-wise predictions. To address this issue, we introduce a simple yet effective pooling operation called the Haar Wavelet-based Downsampling (HWD) module. This module can be easily integrated into CNNs to enhance the performance of semantic segmentation models. The core idea of HWD is to apply Haar wavelet transform for reducing the spatial resolution of feature maps while preserving as much information as possible. Furthermore, to investigate the benefits of HWD, we propose a novel metric, named as feature entropy index (FEI), which measures the degree of information uncertainty after downsampling in CNNs. Specifically, the FEI can be used to indicate the ability of downsampling methods to preserve essential information in semantic segmentation. Our comprehensive experiments demonstrate that the proposed HWD module could (1) effectively improve the segmentation performance across different modality image datasets with various CNN architectures, and (2) efficiently reduce information uncertainty compared to the conventional downsampling methods. Our implementation are available at <https://github.com/apple1986/HWD>.

1. Introduction

Semantic segmentation is a fundamental task in computer vision that involves assigning a label to each pixel of an input image [1]. It has many real-world applications, including vehicle navigation [2] and medical image analysis [3]. The development of deep convolutional neural networks (DCNNs), such as AlexNet [4] and ResNet [5], has led to significant progress in semantic segmentation. These networks consist of multiple layers of convolutions, normalizations, and activation functions. As a basic component of DCNNs, the downsampling operation is frequently applied to alter the resolution of inputs, helping reduce computation overhead and enlarge the receptive field for neural networks.

Deep convolutional neural networks (DCNNs) commonly employ standard downsampling operations, such as max pooling, average pooling, and strided convolution, which may lead to information loss. The lost information, like boundary and texture, is likely essential for

semantic segmentation [6,7]. To alleviate this problem, several approaches have been proposed, which can generally be summarized as follows: (1) feeding more information by skip connection to a decoder sub-network like U-Net [8], LCU-Net [9], CENet [10], LinkNet [11], and RefineNet [12]. As with U-Net, the downsampled feature maps from the encoder are passed to the corresponding stage of the decoder via skip connections. In RefineNet, the information at different stages of the decoder is fully exploited and fused with skip connections to obtain a high-resolution prediction; (2) extracting multi-scale feature maps with spatial pyramid pooling or dilated convolution to the fusion module, such as DeepLab [13], PSPNet [14], PCPLP-Net [15], BiSenet [16], and ICNet [17]. Specifically, in DeepLab, the encoder module converts multi-scale information by the proposed atrous spatial pyramid pooling (ASPP) block to refine segmentation results by gradually recovering the spatial information. And for the PSPNet, a pyramid pooling module was proposed to harvest different scale representations for compensating losing information; (3) providing multi-modality images to the encoder,

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<https://doi.org/10.1016/j.patcog.2023.109819>

Received 4 August 2022; Received in revised form 15 June 2023; Accepted 12 July 2023

Available online 13 July 2023

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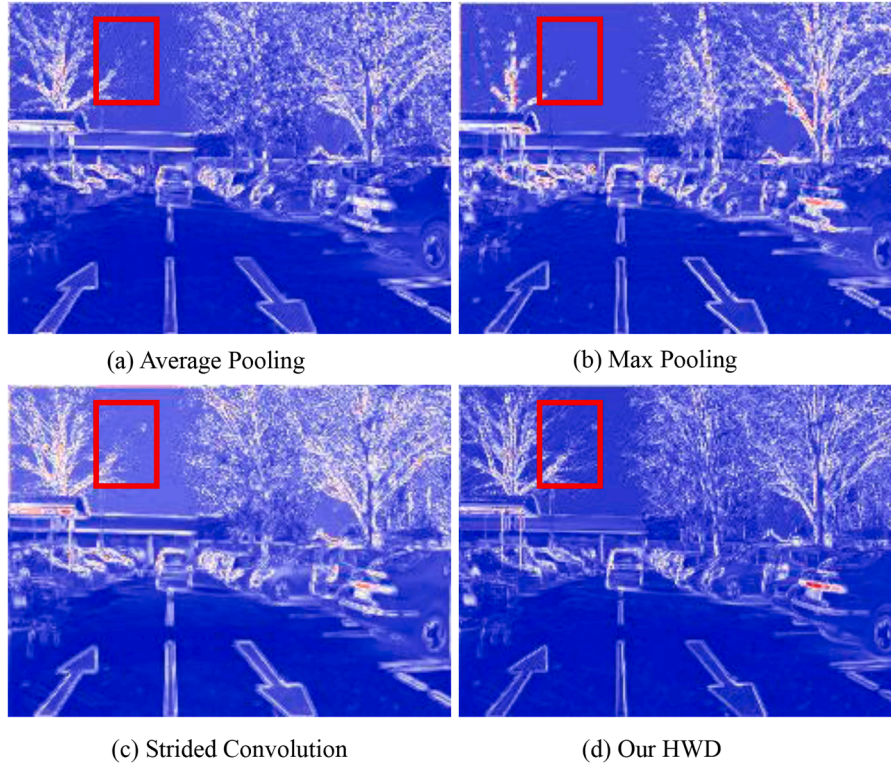


Fig. 1. Downsampling examples of average pooling, max pooling, strided convolution and the HWD in DeepLabv3+ [13]. Compared with conventional downsampling methods, the feature after HWD preserves more boundary, texture and detail information, as marked by the four red squares where tree branches are better preserved in (d). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

like DiSegNet [18], MMADT [19], CANet [20] and CCFFNet [21]. Due to the complementary properties of images from different modalities, it could also mitigate the effect of information loss after downsampling; (4) increasing prior information. A contour-enhanced attention module was proposed in [22], aiming to extract boundary and shape cues from CT images for refining the segmented areas. In [23], superpixels are used to help guide local and global consistency by considering rule-based appearance information. Besides, ensemble adversarial attacks are introduced in [24] for generating highly diverse adversarial example, aiming to enhance the diversity of perturbation over class distribution. In [25] and [26], category prior information is brought into CNNs to increase robustness of DCNNs.

In a nutshell, the main purpose of these methods is to help establish a good relationship between downsampled features and segmentation labels by providing more learned information or features based on various strategies, such as multi-scale, prior guidance, and multi-modality.

Although the previous works could partly alleviate the problem of information loss after downsampling and help to extract representative feature maps by aggregating more information. Yet, the main problem is that simple aggregation of features (e.g., summation or concatenation) may introduce unrelated information into the network, which hinders for discriminative feature learning [27,28]. In other words, conventional downsampling methods in DCNNs lead to information loss. The lost information, such as boundary, scale, and texture, is important for object segmentation, especially for small-scale objects or the objects obscured between the background and foreground [7,29]. Despite the incorporation of various techniques such as skip connections, multi-scale features, and prior knowledge, recovering the essential lost information through subsequent layers remains a significant challenge. Hence, *our first question* is that if regular downsampling methods would cause information loss, can we design an information-preserving downsampling module to keep information as much as possible in

DCNNs for semantic segmentation?

Inspired by the lossless information transformation methods [30], we introduce the Haar wavelet transform into our downsampling module. Theories and practice of Haar wavelet transform have been extensively studied [31]. Early works of Haar wavelet transform in image processing have focused on image decomposition [32], compression [33], denoising [34], reconstruction [35] etc. The main advantage of the Haar wavelet is its multi-scale lossless fast signal decomposition. In this paper, we make use of this advantage from Haar wavelet transform, and propose Haar wavelet downsampling (HWD) module, a new downsampling method for segmentation task. Our main motivation is if the information of an image could be preserved as much as possible at the encoder part of deep neural network, it will be easier to extract more discriminative features which may help improve the segmentation performance. Hence, in contrast to conventional downsampling methods that directly decrease the spatial resolution of feature maps with losing information, we propose an alternative approach. Firstly, it explicitly increase the channel number of feature maps and reduce its resolution using Haar wavelet transform without loss of information. Then, we employ convolution operations for representative feature learning to filter redundant information. Figure 1 shows several downsampling examples using average pooling, max pooling, strided convolution and the proposed HWD in DeepLabv3+ [13]. We can see the HWD could preserve more detail information comparing to other three downsampling methods.

With the integration of HWD in DCNNs, *the second question* arises: how to confirm or measure the amount of essential features which have been learned by its subsequent convolution layers? Here, the essential features mean that they could help infer the DCNNs to generate better predictions comparing with the corresponding ground truth. In other words, if extracted features are representative, the neural network will have more confidence (certainty) to predict the desired results, suggesting that the uncertainty of information is relatively low. Previous

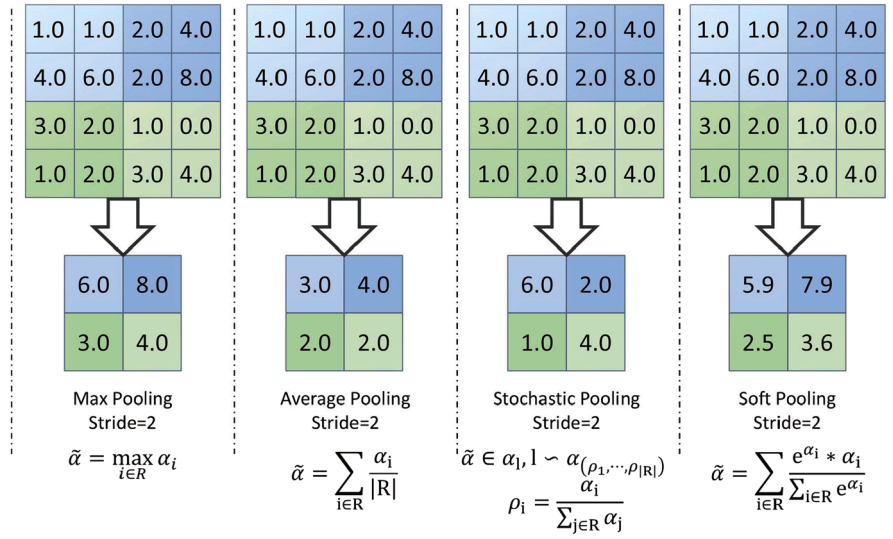


Fig. 2. Illustration of four pooling methods.

studies have widely utilized the concept of information entropy [36] to assess the degree of uncertainty or randomness present in signals or images in a communication system [37]. Considering a neural network as a communication system, the primary purpose of this system in segmentation tasks is to reduce uncertainty between input images and their corresponding semantic labels [38,39]. Inspired by the concept of information entropy to evaluate uncertainty of information, we proposed a novel metric called Feature Entropy Index (FEI) to quantify the uncertainty between the feature and prediction output in DCNNs. In particular, we could employ the FEI to estimate the degree of uncertainty of the downsampled features, which could reflect the importance of the features relative to the ground truth.

In summary, our main contributions are as follows:

- (1) We propose a novel wavelet-based downsampling module (HWD) for CNNs. To the best of our knowledge, our method is the first attempt to explore the feasibility by prohibiting (impeding) information loss in the downsampling stage of DCNNs for the semantic segmentation task.
- (2) We explore the measurement of information uncertainty across layers in CNNs, and propose a novel index (FEI) to evaluate the information uncertainty or feature importance between the downsampled feature maps and the prediction results.
- (3) The proposed HWD can directly replace the strided convolution or pooling layer without significantly increasing the computational overhead, and can be easily integrated into current segmentation architectures. Extensive experiments demonstrate the effectiveness of the HWD module in comparison with seven state-of-the-art (SOTA) segmentation methods.

The rest of this paper is organized as follows: Section 2 reviews the related work of downsampling in DCNNs. Section 3 describes the proposed HWD module and the definition of FEI. Experiments and results are presented in Section 4. Section 5 gives the discussion and conclusion of this study and points out the limitations and future works.

2. Related work

2.1. Wavelet transform in CNNs

Several studies have explored the utilization of wavelet transform in CNNs to improve feature representation in various tasks such as classification, super-resolution, denoising, and more. In [40], a wavelet CNN

that integrates multi-scale resolution analysis into CNNs was presented and applied to the task of texture classification and image annotation, yielding superior accuracy compared to prior approaches. In [41], a multi-level wavelet CNN (MWCNN) model was proposed for image restoration, aiming to strike a better balance between receptive field size and computational efficiency. In [42], the MWCNN was utilized as a denoiser prior for restoring blurred images corrupted by Cauchy noise. Similarly, a wavelet-based CNN (Wavelet-SRNet) was introduced for multi-scale face super-resolution, where wavelet coefficients of high-resolution feature maps were learned before reconstructing high-resolution images [43]. In [44], a wavelet-like transform was integrated into CNN for image compression, using an update-first lifting scheme to support multi-resolution analysis. These studies leverage the benefits of feature learning from CNNs and multi-scale resolution analysis offered by wavelet techniques. In contrast to these studies, we investigate the Haar wavelet transform as an alternative of the downsampling method in CNNs, and focus on the image segmentation task.

2.2. Downsampling methods for feature maps

Downsampling operations offer inherent advantages, leading to their widespread utilization in diverse tasks, including ordinal classification [26], deep face recognition [45], semantic re-ranking systems [25], and autonomous driving [24]. Numerous downsampling methods have been proposed to reduce the spatial resolution of feature maps, which could decrease the computational requirements of CNNs while increasing the receptive field of subsequent convolutions such as ResNet [5] and U-Net [8]. Currently, there are two types of downsampling methods that are frequently used in CNNs, namely the pooling operation and the strided convolution operation. Figure 2 illustrates four types of pooling methods. The pooling operations are based on a neighborhood approach and have no additional learning parameters. The max-pooling and the average pooling are the two main downsampling methods adopted in many segmentation architectures, like FCN, U-Net and PSPNet. There are also some works on adaptive pooling, like stochastic pooling [46] and softpool [47]. In [48], a wavelet pooling for CNNs has been introduced. This method decomposes features into two-level decomposition, and discards the first-level subbands to reduce feature dimensions, which shows superiority in addressing the overfitting problem in image classification task. The second category of downsampling uses strided convolution operation to reduce the size of feature maps. It aims to preserve perceptually important details and align local image features in a learning manner [49,50]. For example, Chen et al. [51] proposed a

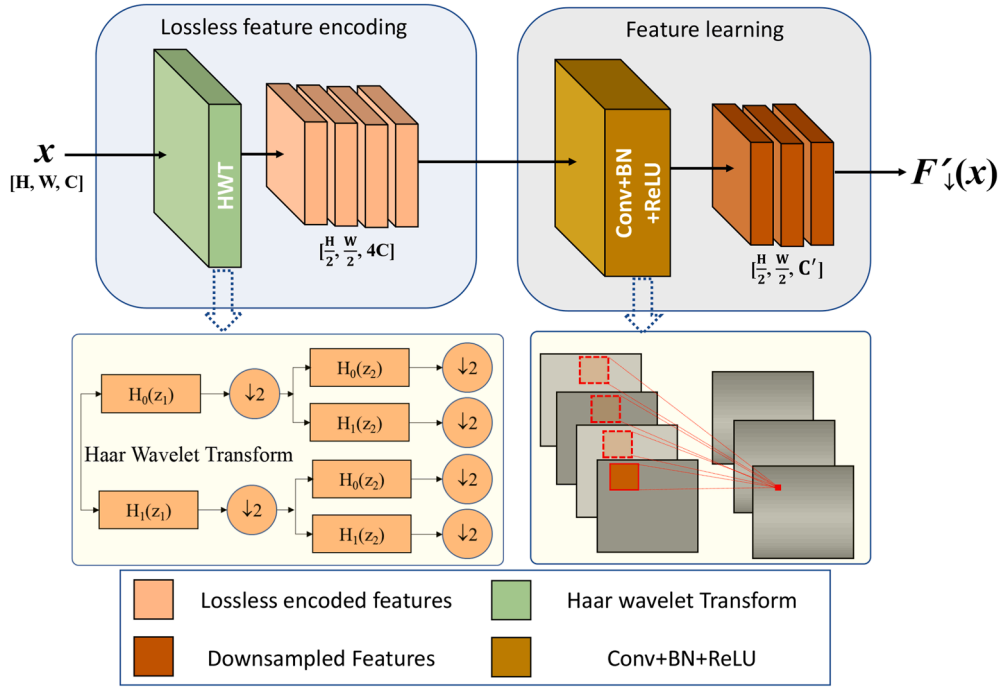


Fig. 3. The architecture of the proposed HWD module which consists of two main blocks: lossless feature encoding block and feature representation learning block. Note that the channel number of feature maps can be adjusted by the representation learning block.

convolutional block for learning fractional downsampling. If we consider the downsampling operation as information encoding processing, the above downsampling (pooling or strided-convolution) methods will result in information loss. The lost information, such as boundary, scale, and texture, plays an essential role in semantic segmentation. In other words, subsequent layers will have a better ability to learn representative features if more information is preserved after downsampling. Therefore, unlike previous studies, we proposed a simple yet effective downsampling module HWD in this paper, aiming to information preserving while reducing spatial resolution of feature maps. Any kind of downsampling operation, like strided convolution or max pooling, can be directly replaced with HWD at a minimal cost.

2.3. Image quality assessment

Extensive research has been conducted in the field of Image Quality Assessment (IQA) to evaluate image distortion and human observers' perception. Based on the availability of an original reference image, IQA methods can be broadly categorized into two approaches: full-reference (FR) and no-reference (NR). As FR IQA, peak signal-to-noise ratio, mean square error and structural similarity index measure (SSIM) [52] have become commonly used metrics. In particular, the SSIM, which is designed by modeling a combination of correction loss, luminance distortion and contrast distortion, is widely accepted because of its reasonable correlation with the perception of the human visual system (HVS). Several variants based on SSIM have been proposed, such as the Multi-Scale Structural Similarity Index (MSSIM) [53], which incorporates multi-scale spatial information, and the Information-Weighted Structural Similarity Index (IWSSIM) [54], which integrates information-weighted entropy. Furthermore, the researchers found that the HVS is more sensitive to visual information [55,56]. The Gradient and Structure Similarity Model (GSM) [55] integrates gradient and luminance information to provide a comprehensive assessment of image quality. Similarly, the Feature Similarity Index Measure (FSIM) [56] employs phase congruency and gradient magnitude to assess image quality.

With the development of deep learning, some NR metrics based on

deep neural networks have been emerged for IQA. For instance, Inception score (IS) [57] was found to evaluate the image quality in generative adversarial networks. It directly applied a pretrained Inception model for generated image to get the conditional label distribution based on the assumption that high quality image should contain meaningful objects. The Fréchet Inception Distance (FID) [58] is a metric that calculates the distance between feature vectors calculated for real and generated images.

In summary, the majority of IQA metrics are commonly employed to evaluate the quality of the original images. This study focuses on the development of a metric for assessing the quality of downsampled feature maps. The metric aims to quantify the degree of information uncertainty or feature importance by comparing the downsampled feature maps with the prediction results.

3. Methods

As illustrated in Fig. 3, the proposed HWD module comprises two block: (1) the lossless feature encoding block and (2) the feature representation learning block. The lossless feature encoding block is responsible for transforming features and reducing spatial resolution. To accomplish this, we utilize the Haar wavelet transform, a method that efficiently decreases the resolution of feature maps while retaining all information. The representation learning block consists of a standard convolution layer, batch normalization, and a ReLU activation layer. It is employed to extract discriminative features. Each block will be thoroughly explained in the following subsection.

3.1. Lossless feature encoding block

The lossless feature encoding block utilizes a Haar wavelet transform layer to effectively reduce the spatial resolution of feature maps while preserving all information. The Haar wavelet transform is a widely recognized, compact, dyadic, and orthonormal transform that finds extensive application in image coding, edge extraction, and binary logic design [31]. The wavelet basis function and scale function for the 1-stage, one-dimensional Haar transform can be defined as follows:

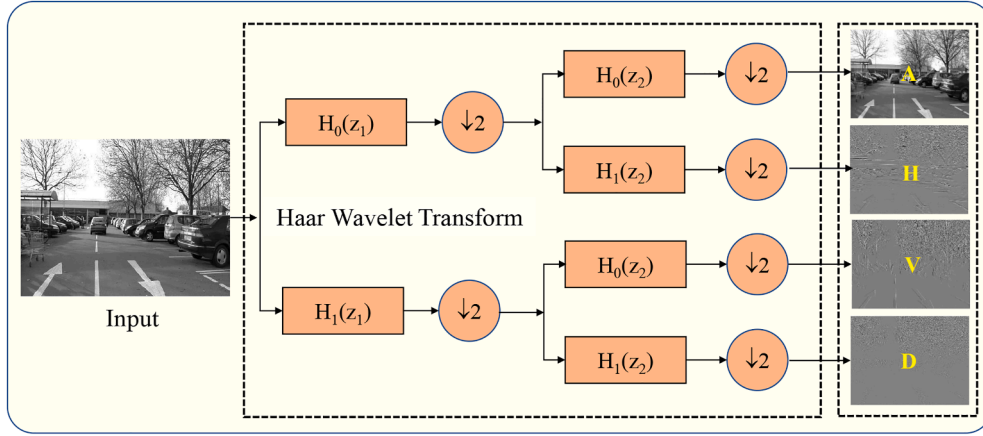


Fig. 4. The illustration of wavelet transform method, which encodes the input image into four components with reduced spatial resolution comparing to the input.

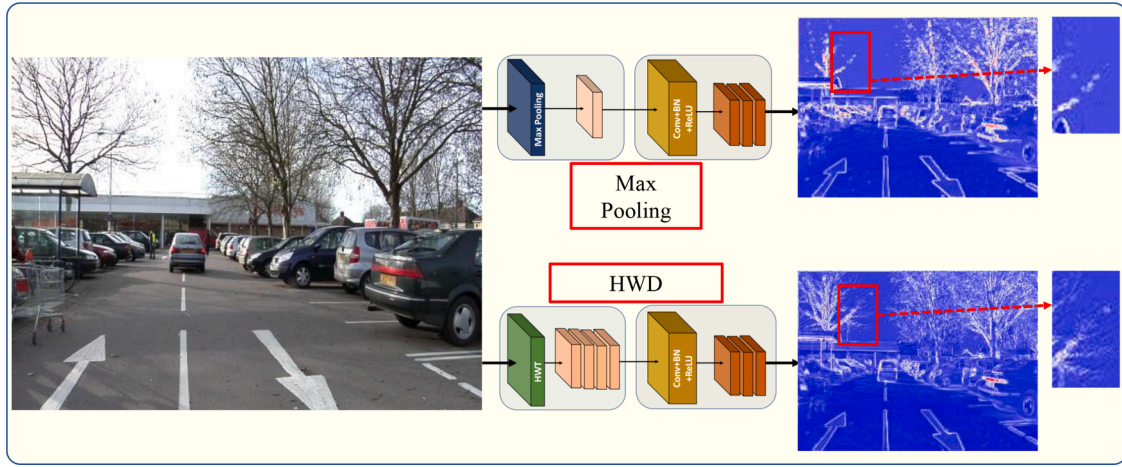


Fig. 5. Illustration of HWD and max pooling downsampling methods for a RGB image. Top row: downsampled feature with max pooling operation. Bottom row: feature extraction with HWD. (Note that the displayed feature is obtained by maximizing all its downsampled feature maps along channel direction.).

$$\begin{cases} \phi_1(x) = \frac{1}{\sqrt{2}}\phi_{1,0}(x) + \frac{1}{\sqrt{2}}\phi_{1,1}(x) \\ \psi_1(x) = \frac{1}{\sqrt{2}}\phi_{1,0}(x) - \frac{1}{\sqrt{2}}\phi_{1,1}(x). \end{cases} \quad (1)$$

Here, $\phi_{j,k}(x)$ is defined as:

$$\phi_{j,k}(x) = \sqrt{2^j} \phi(2^j x - k), k = 0, 1, \dots, 2^j - 1. \quad (2)$$

In this context, the parameters j and k are denoted as the stage (or scale in the image processing domain) and the order (or direction for 2D image) of the Haar basis function, respectively. Furthermore, $\phi_{0,0}(x)$ is defined as:

$$\phi_{0,0}(x) = \phi_0(x) = \begin{cases} 0, & x < 0 \\ 1, & 0 \leq x < 1, \\ 0, & x \geq 1 \end{cases} \quad (3)$$

Therefore, the 1-stage Haar transform can be expressed using the 0-stage Haar basis functions:

$$\begin{cases} \phi_1(x) = \phi_0(2x) + \phi_0(2x - 1) \\ \psi_1(x) = \phi_0(2x) - \phi_0(2x - 1). \end{cases} \quad (4)$$

This implies that a signal of length L can be divided into two parts of length $\frac{L}{2}$, which can be interpreted as low-pass and high-pass decomposition filters, respectively. When applying the Haar wavelet transform

to a two-dimensional signal, such as a grayscale image, yields four components, each with half the spatial resolution of the original signal.

Figure 4 depicts the decomposition process of an image with the resolution of $H \times W$ using the Haar Wavelet transform. Here, H_0 and H_1 represent the low-pass and high-pass decomposition filters, respectively. These filters are employed to extract the approximate and high-frequency information from an image, respectively. The symbol $\downarrow 2$ denotes downsampling applied to the approximate and detail components. The Haar wavelet transform generates four components: the approximate (low-frequency) component (A), as well as the detail components (high frequency) in the horizontal (H), vertical (V), and diagonal (D) directions. Each component has a size of $\frac{H}{2} \times \frac{W}{2}$. It should be noted that the resolution of each component is reduced to $\frac{H}{2} \times \frac{W}{2}$, whereas the channel number of feature maps has quadrupled. In other words, the Haar Wavelet transform can encode partial information from the spatial dimension into the channel dimension without any loss of information. Consequently, the subsequent layers of DCNNs can extract representative features from the transformed components, which have reduced spatial resolution.

3.2. The representation feature learning block

The feature representation learning block consists of a standard 1×1 convolution layer, a batch normalization layer, and a ReLU activation function. In this block, standard convolution is employed to adjust the

Table 1

Comparison of parameters and FLOPs for three downsampling methods.

Module	Parameters	FLOPs
HWD	$4C^2$	$2HWC^2 + 3.75HWC$
Average Pooling	0	HWC
Strided Convolution	$9C^2$	$4.5HWC^2 - 0.25HWC$

channel number of feature maps. This block serves two main purposes: (1) to adjust the channel numbers of feature maps to align with the subsequent layer, and (2) to filter redundant information as much as possible, enabling subsequent layers to learn representative features more effectively. This block allows the HWD module to be substituted with various downsampling methods, such as max pooling or strided convolution. In Fig. 5, we compare outputs between the proposed HWD module and max pooling operation. We can see that the output feature map have more details from the HWD compared to the max pooling.

In summary, the proposed HWD module consists of two main blocks. The first block focuses on reducing the spatial resolution of feature maps using Haar transform, while the second block aims to filter redundant information through standard 1×1 convolution, batch normalization, and ReLU operations. Assuming the input size of feature maps is $H \times W \times C$, and the size of downsampled feature maps is $\frac{H}{2} \times \frac{W}{2} \times C$. When compared to traditional 3×3 convolution with stride 2 and average pooling as downsampling, the total parameters and computational overhead are presented in Table 1. It can be seen that average pooling outperforms in terms of parameters and Floating Point Operations (FLOPs). In comparison, our HWD module requires fewer than twice the parameters of strided convolution. Additionally, when the channel number C is greater than one, the computation overhead of strided convolution exceeds that of HWD. As a result, the HWD module provides a trade-off between average pooling and strided convolution in parameters and FLOPs.

3.3. The feature entropy index

To evaluate the uncertainty of the feature maps after downsampling, we introduce a novel metric called the feature entropy index (FEI), which is inspired by the concept of information entropy. The definition of feature entropy is given by:

$$FEI = - \sum_{m=1}^M \left[\left(\frac{1}{C} \sum_{c=1}^C F \right) \log_{10} P \right], \quad (5)$$

where F is feature maps after downsampling which has total C number of channels with the spatial resolution $H \times W$ after bilinear interpolation, and P is the final prediction result with the size of $H \times W$ from segmentation model. Each pixel value from P means the maximum prediction probability for each class. M means the total pixel number of P , which is equal to $H \times W$. The FEI enables us to quantify the uncertainty of feature maps generated by convolutional neural networks (CNNs). A smaller FEI value indicates reduced uncertainty in the feature maps, implying that they offer more information for uncertainty reduction. When FEI reaches zero, it signifies that the model can accurately infer the segmentation using the current feature maps.

4. Experimental results

To evaluate the effectiveness of the HWD, extensive experiments are conducted on three different image modalities, including natural image, Computed Tomography (CT), and Micro-optical sectioning tomography (MOST) images. In this section, we provide a brief introduction to the datasets and discuss the implementation details. Subsequently, we report the segmentation results and compare them with state-of-the-art (SOTA) methods on these three datasets.

4.1. Datasets

Camvid dataset: This dataset, originally designed for street scene understanding, consists of 701 images. The dataset is divided into training (367 images), validation (101 images), and testing (233 images) sets. It includes a total of 32 semantic classes. Following the previous work [59], we focus on 11 classes: sky, building (BLD), pole (PoL), road (Roa), pavement (Pav), tree (Tre), traffic sign (Sig), fence (Fen), car, pedestrian (Ped), and bike (Bik). The original resolution of images is 720×960 . We resize them into 360×480 for easy comparison with prior work, and take the intersection of union (IoU) to evaluate the performance of HWD.

Synapse multi-organ segmentation dataset (Synapse): This dataset comprises 30 abdominal CT scans, consisting of a total of 3779 axial contrast-enhanced abdominal clinical CT images. Following the data splitting strategy in [60], we use 18 cases for training and the remaining 12 cases for validation. The performance evaluation is based on the mean Dice Similarity Coefficient (mDSC) and the mean Hausdorff Distance (mHD) for eight abdominal organs: aorta, gallbladder (Gall), spleen, left kidney (KidL), right kidney (KidR), liver, pancreas, spleen, and stomach (Stom).

Micro-optical sectioning tomography dataset (MOST) [61]: The voxel size of this dataset is $0.35 \times 0.35 \times 1 \mu\text{m}$. The soma and vessels are labeled by two human experts. For our experiments, we randomly split the dataset into 1100 training images and 100 testing images, all with the same resolution, following a previous study [62,63].

4.2. Implementation details

All experiments were conducted on a workstation running Ubuntu 18.04 LTS, utilizing Python 3.8 and PyTorch 1.8.0. The workstation was equipped with two Nvidia 3090 GPUs with 24GB memory and an Intel i9 CPU. Real-time augmentation strategies, such as random flipping and rotations, were employed to enhance data diversity for all training instances. The models were trained using the Adam optimizer with a initial learning rate of 5×10^{-4} and a weight decay of 1×10^{-4} . The ‘‘poly’’ learning rate schedule proposed in [13] was also utilized, with a power of 0.9. Each iteration had a batch size of 8. The training epochs were set to 350 for Camvid, 350 for Synapse, and 200 for MOST.

The sum of cross entropy and Generalized Dice loss is defined as the loss function during the training [64], which is calculated below:

$$Loss(g, p) = -\frac{1}{N} \left(\sum_{n=1}^N \sum_{c=1}^C g_i^c \log p_i^c \right) + \left(1 - \frac{2 \sum_{n=1}^N \sum_{c=1}^C g_i^c p_i^c}{\sum_{n=1}^N \sum_{c=1}^C (g_i^c)^2 + \sum_{n=1}^N \sum_{c=1}^C (p_i^c)^2} \right), \quad (6)$$

where g , p , N and C denote the ground truth, prediction results, the number of image elements, and label classes, respectively. The first and second items refer to the definitions of cross-entropy and generalized Dice loss, respectively. According to Chen et al. [60], all 3D volume datasets are trained slice by slice, and the predicted 2D slices are then stacked together to form a 3D prediction for evaluation on Synapse. For the other two datasets, we evaluate the segmentation performance using 2D predictions. The evaluation metrics, namely intersection-over-union (IoU), Dice similarity coefficient (DSC), and Hausdorff distance (HD), are used to evaluate the performance. The definitions of each metric are as follows:

$$IoU = \frac{TP}{TP + FP + FN}, \quad (7)$$

$$DSC = \frac{2TP}{2TP + FP + FN}, \quad (8)$$

Table 2

Performance of seven SOTA segmentation architecture with three types of ResNet as backbones on Camvid dataset.

Method	Backbone	mIoU↑	Sky	BLD	Pol	Roa	Pav	Tre	Sig	Fen	Car	Ped	Bik	Time(ms)
PSPNet	R-18	57.50	90.25	77.71	18.82	93.46	79.63	71.77	29.10	20.09	77.76	40.23	33.73	2.18
	R-34	60.23	90.35	77.87	20.86	93.59	79.06	71.56	33.95	26.47	81.92	43.38	43.55	2.73
	R-50	58.63	90.88	78.23	20.73	93.95	80.22	72.85	33.53	17.66	80.69	38.45	37.76	3.26
LinkNet	R-18	60.30	91.71	78.01	24.72	94.35	81.35	70.55	31.25	23.63	80.26	43.21	44.29	3.88
	R-34	61.78	91.73	78.85	25.91	94.22	81.31	71.05	35.44	23.11	81.76	45.56	50.61	5.35
	R-50	63.04	92.23	80.22	28.15	94.22	81.12	72.90	36.02	31.14	82.03	46.08	49.37	6.39
FPN	R-18	62.71	91.28	79.94	27.94	95.04	82.92	71.67	33.13	27.38	84.86	45.52	50.13	3.75
	R-34	64.36	91.40	80.74	26.46	94.79	82.43	73.57	36.26	34.38	84.81	49.88	53.19	5.24
	R-50	63.82	91.44	80.13	26.71	95.16	83.83	72.47	35.63	29.66	84.37	49.55	53.03	6.25
PAN	R-18	61.47	90.64	79.72	25.33	94.66	81.95	70.88	34.08	27.58	82.73	44.96	43.66	4.41
	R-34	62.94	90.73	80.70	27.19	94.84	82.41	72.47	37.31	28.40	85.79	46.24	46.31	5.82
	R-50	60.66	89.86	78.66	25.70	94.38	81.15	71.93	30.85	22.52	82.75	46.70	42.72	6.98
DeepLabv3+	R-18	61.27	90.72	79.16	24.48	94.41	81.60	71.93	34.13	20.79	83.73	47.42	45.55	3.40
	R-34	61.39	91.23	79.48	24.76	94.72	82.59	72.29	34.78	22.36	83.32	46.21	43.59	4.88
	R-50	61.74	91.19	79.43	26.60	94.64	81.69	72.31	33.36	20.82	83.21	49.12	46.79	5.93
U-Net	R-18	62.54	91.91	79.20	27.96	94.51	82.51	71.59	33.34	27.47	84.17	47.75	47.54	3.64
	R-34	64.61	92.23	80.55	30.20	95.28	84.16	73.00	39.12	28.24	84.24	51.35	52.36	5.16
	R-50	64.33	91.93	80.68	29.38	94.83	82.88	73.69	37.56	28.85	83.54	52.54	51.77	6.28
Unet+	R-18	63.50	92.18	80.17	28.34	94.66	82.61	72.04	37.21	30.33	84.40	49.85	46.66	5.38
	R-34	64.89	92.11	80.94	30.99	95.06	83.90	73.55	38.84	26.96	85.82	53.59	52.00	6.91
	R-50	64.86	92.24	80.93	29.81	94.96	82.75	74.38	40.61	28.74	83.18	53.25	52.63	8.24
Mean	/	62.22	91.34	79.59	26.24	94.56	82.00	72.31	35.02	26.03	83.11	47.18	47.01	5.05

Table 3

Performance of seven SOTA segmentation architecture with three types of ResNet equipped with HWD module as backbones on Camvid dataset.

Method	Backbone	mIoU↑	Sky	BLD	Pol	Roa	Pav	Tre	Sig	Fen	Car	Ped	Bik	Time(ms)
PSPNet_HWD	R-18	58.83	90.38	77.38	20.47	94.18	80.12	72.46	29.42	28.50	79.49	39.67	35.02	2.86
	R-34	61.15	90.53	78.99	21.93	94.57	81.58	72.77	36.32	27.10	81.52	43.40	43.97	3.44
	R-50	60.08	90.91	78.32	22.43	94.34	80.55	73.61	32.14	25.41	81.34	42.13	39.74	3.90
LinkNet_HWD	R-18	63.02	91.91	80.32	28.98	94.77	82.74	71.34	35.33	29.24	83.67	49.06	45.88	4.45
	R-34	64.07	92.28	79.99	29.87	94.34	81.61	72.05	36.13	31.94	85.70	49.44	51.46	6.02
	R-50	64.63	92.20	81.09	30.43	93.91	80.83	74.34	39.56	31.22	82.69	50.75	53.93	7.15
FPN_HWD	R-18	64.66	91.57	80.40	28.90	95.14	82.87	73.36	35.95	34.07	84.53	50.00	54.50	4.44
	R-34	65.15	91.49	81.33	28.47	95.20	83.66	73.77	38.11	33.47	86.51	49.98	54.70	6.04
	R-50	64.74	91.46	81.44	28.39	94.80	82.54	73.59	38.86	31.22	85.80	50.56	53.51	7.04
PAN_HWD	R-18	62.69	90.93	80.00	25.86	94.63	82.18	72.20	33.19	32.69	83.68	47.41	46.85	5.02
	R-34	64.11	90.98	80.28	28.18	95.39	84.10	72.13	37.49	31.08	85.42	50.44	49.71	6.54
	R-50	61.10	91.19	78.69	24.99	94.13	79.97	70.39	34.05	23.53	82.82	45.84	46.54	7.86
DeepLabv3+_HWD	R-18	62.75	91.33	80.24	26.54	94.42	80.64	72.06	35.49	30.26	84.53	49.09	45.61	4.00
	R-34	63.07	90.99	80.71	28.93	94.43	81.14	72.96	36.17	23.78	85.25	49.56	49.88	5.55
	R-50	62.31	91.56	80.02	27.80	94.20	80.78	73.19	33.02	21.84	82.60	50.99	49.42	6.62
U-Net_HWD	R-18	64.31	91.83	80.51	30.40	94.93	83.34	71.87	36.54	37.14	82.44	50.13	48.27	4.29
	R-34	65.62	92.23	80.93	30.91	95.41	84.68	73.52	38.98	37.97	83.43	53.83	49.92	5.84
	R-50	64.28	92.35	80.82	29.97	94.49	82.23	74.07	39.03	29.51	83.51	52.87	48.25	6.87
Unet+_HWD	R-18	65.98	92.30	81.48	32.30	95.04	83.40	73.76	38.95	34.48	86.27	53.83	54.01	6.03
	R-34	66.10	92.48	81.63	31.61	95.66	84.77	73.93	41.67	32.08	86.33	52.67	54.30	7.87
	R-50	66.48	92.42	82.11	33.09	94.94	83.23	74.43	45.55	37.98	83.61	52.39	51.56	8.98
Mean	/	63.58	91.59	80.32	28.12	94.71	82.24	72.94	36.76	30.69	83.86	49.24	48.91	5.75

$$HD = \max_{x \in X} \min_{y \in Y} \|x - y\|_2, \quad (9)$$

where TP , FP , and FN , x and y mean true positive, false positive, false negative, and two points from two finite points sets X and Y .

Considering that ResNets have become the prevailing backbone in numerous segmentation architectures, we employ ResNet-18, ResNet-34, and ResNet-50 as the backbone for feature learning in this study. Additionally, in the Appendix, we present the segmentation performance achieved by adopting MobileNetv2 as the backbone, as well as other SOAT segmentation architectures based on convolution or transformer operations. Basically, a standard ResNet comprises a stem with two downsampling operations, followed by four stage blocks that consist of convolution layers and residual connections. Our primary objective is to maximize the retention of information at the stem of the ResNet. Therefore, we straightforwardly substitute the initial convolution layer (with a stride of 2) and the first max pooling layer in the stem of ResNet with HWD, while keeping all the subsequent layers identical to the original ResNet architecture. It is worth noting that all models are trained from scratch.

4.3. Experimental results on Camvid dataset

(1) Comparison with SOTA methods

We evaluate our method on seven state-of-the-art (SOTA) segmentation architectures: DeepLabv3+, PSPNet, LinkNet, FPN, PAN, U-Net, and Unet++. ResNet-18 (R-18), ResNet-34 (R-34), and ResNet-50 (R-50) are selected as the backbones for feature learning in each architecture. Further details about the training loss can be found in the Appendix.

Evaluation: Tables 2 and 3 present a comparison of our plug-in-plug-off HWD module's effectiveness against the baselines in terms of mean IoU. The SOTA segmentation architectures equipped with the HWD module exhibit a performance improvement of 1~2% for each respective architecture. Specifically, our method demonstrates its effectiveness on this dataset by improving the mIoU of all 21 models by 1.21%. Particularly, our method significantly improves the performance of small-scale objects, such as pedestrian (+2.06%), bicycle (+1.9%), fence (+4.66%), and sign symbol (+1.74%). This suggests that segmenting large-scale objects is relatively easy, even when downsampling

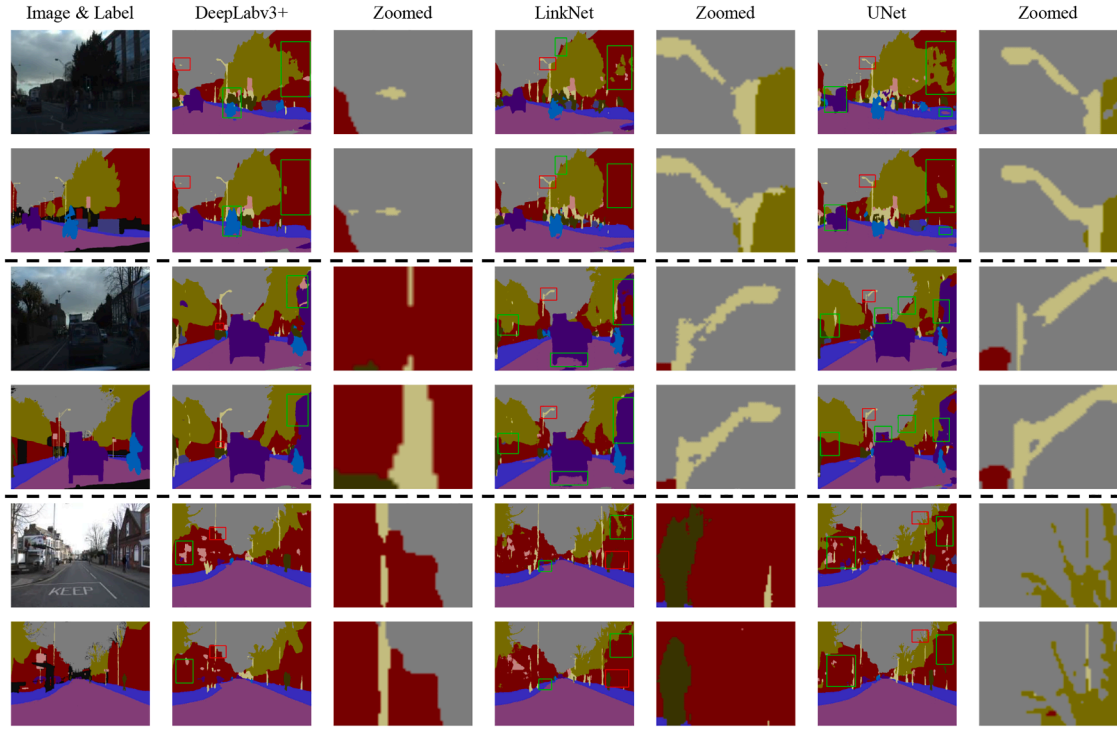


Fig. 6. Visualized segmentation results on Camvid test set. The first column contains the input image and its corresponding ground truth. The second, fourth, and sixth columns display the outputs of DeepLabv3+, LinkNet, and U-Net, respectively, with ResNet-34 as the backbone. The third, fifth, and seventh columns show zoomed-in images that correspond to the regions highlighted in red boxes. More specifically, the even rows represent the segmentation results obtained with HWD, while the odd rows correspond to the results using the original downsampling in ResNet-34. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 4

Results of SSIM (%), PSNR (dB), FEI and FEI on boundary loop (testing on Camvid testing dataset).

Model	Backbone	SSIM↑	PSNR↑	FEI↓	FEI_B↓	Model	SSIM↑	PSNR↑	FEI↓	FEI_B↓
DeepLabv3+	R-18	70.66	9.29	219.70	176.32	DeepLabv3+_HWD	78.42	11.07	94.63	73.22
	R-34	70.92	8.95	240.39	191.90		76.23	10.26	90.10	70.00
	R-50	71.87	8.74	266.47	212.83		75.38	10.87	121.74	92.44
FPN	R-18	67.33	7.71	244.62	197.79	FPN_HWD	79.35	10.94	100.22	78.97
	R-34	67.84	7.49	255.78	205.79		79.30	10.84	96.98	77.07
	R-50	71.67	8.46	253.36	204.54		78.94	11.33	105.83	82.89
Linknet	R-18	72.12	11.36	200.28	152.82	Linknet_HWD	78.71	10.84	92.37	70.15
	R-34	72.26	10.63	227.73	172.98		79.80	11.03	98.56	74.90
	R-50	71.51	9.83	219.57	170.36		80.33	11.91	107.27	82.81
PAN	R-18	70.00	8.56	237.82	191.30	PAN_HWD	77.44	10.54	95.52	73.29
	R-34	70.39	8.54	241.81	193.92		77.83	10.77	100.73	78.02
	R-50	72.31	9.05	262.54	206.88		78.85	11.30	120.69	92.36
PSPNet	R-18	68.36	7.86	377.00	284.23	PSPNet_HWD	77.93	10.74	153.56	111.35
	R-34	70.63	8.55	325.80	250.17		78.15	10.84	133.43	101.98
	R-50	73.00	8.71	319.11	247.27		79.08	11.50	134.00	100.24
Unet+	R-18	66.91	7.81	224.88	174.71	Unet+_HWD	76.72	10.20	85.42	65.27
	R-34	66.86	7.51	229.25	181.18		75.22	9.94	89.85	68.27
	R-50	69.39	7.69	234.76	184.64		74.84	10.13	101.74	77.89
U-Net	R-18	67.11	8.07	238.13	182.41	U-Net_HWD	75.64	9.81	94.52	71.72
	R-34	68.45	7.98	239.87	189.82		79.01	10.80	94.59	71.46
	R-50	71.55	8.34	235.51	186.38		77.38	10.59	101.84	77.41
Mean	/	70.05	8.63	252.11	198.01	Mean	77.83	10.77	105.41	80.56

operations may lead to information loss. Additionally, we recorded the prediction time for each model on the CPU. The mean inference time increased by 0.7 ms when using the HWD in the stem of ResNet, which is acceptable considering the improved segmentation performance.

Qualitative results: Figure 6 presents the visualization of the segmentation results using three different architectures (DeepLabv3+, U-Net, and LinkNet) with ResNet-34 (R-34) as the backbone, both with and without the HWD module. The three architectures equipped with the proposed HWD downsampling module exhibit improved performance,

which can be summarized in three aspects: 1) Pure ResNet-34-based methods tend to under-segment objects such as sign-symbol, bicycle, and fence, highlighting the ability of HWD to alleviate the under-segmentation issue. 2) When comparing with and without HWD, our method demonstrates improved results for small-scale objects. 3) Comparing with the original ResNet-based models, the predictions using HWD exhibit smoother boundaries and shapes (e.g., tree, sidewalk, and building) compared to those using conventional downsampling methods. (2) Evaluation on feature effectiveness after downsampling

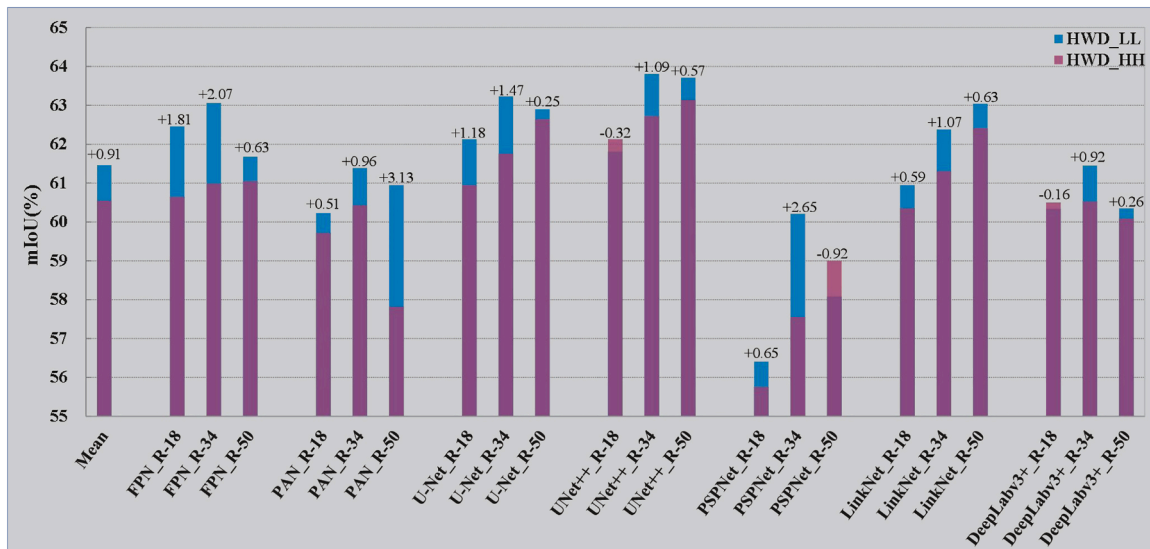


Fig. 7. Comparison the performance by using low frequent component A (denoted as HWD_LL) in HWD and high frequent components (H, V and D, denoted as HWD_HH) in HWD. The “+” in the figure means the improvement performance in terms of mIoU.

Table 5

Mean performance of total 21 segmentation models with ResNet as backbone, by using various number of HWD as downsampling operation on Camvid dataset.

No. of HWD	mIoU↑	Sky	BLD	Pol	Roa	Pav	Tre	Sig	Fen	Car	Ped	Bik
0	62.22	91.34	79.59	26.24	94.56	82.00	72.31	35.02	26.03	83.11	47.18	47.01
2	63.58	91.59	80.32	28.12	94.71	82.24	72.94	36.76	30.69	83.86	49.24	48.91
6	63.40	91.57	80.13	27.89	94.62	81.96	72.86	36.19	30.53	83.94	49.10	48.54
8	63.42	91.47	80.23	27.89	94.69	82.01	72.81	36.44	30.41	83.86	48.95	48.87
10	63.03	0.91	0.80	0.28	0.95	0.83	0.73	0.36	0.29	0.85	0.49	0.48

Table 6

Performance of seven SOTA models without HWD on the Synapse test set.

Methods	Backbone	mDSC↑	mHD↓	Aorta	Gall	KidL	KidR	Liver	Pancreas	Spleen	Stomach
U-Net	R-18	77.37	27.43	85.88	62.08	81.27	72.73	93.91	60.6	90.26	72.24
	R-34	79.21	27.94	87.39	66.81	82.81	76.69	94.02	60.04	90.41	75.51
	R-50	77.84	31.27	88.17	62.39	82.51	78.57	94.03	55.15	88.73	73.18
DeepLabv3+	R-18	75.92	24.23	85.08	61.48	83.32	74.16	93.44	54.46	85.39	70
	R-34	77.37	22.08	85.56	62.69	81.99	75.45	93.17	59.53	87.31	73.27
	R-50	73.87	31.94	83.78	57.23	76.91	71.42	92.91	50.22	87.03	71.43
FPN	R-18	76.56	21.84	85.69	58.88	80.66	73.6	92.81	53.92	90.93	76.02
	R-34	77.89	15.86	86.62	61.42	84.94	76.25	93.32	55.45	90.2	74.94
	R-50	77.08	21.19	85.84	59.56	81.63	78.4	93.55	52.82	89.57	75.28
PSPNet	R-18	74.19	20.65	82.05	58.58	80.86	73.69	92.16	46.22	86.44	73.52
	R-34	75.89	16.43	82.39	62.65	82.38	77.24	93.22	49.54	85.66	74.05
	R-50	76.3	21.3	82.74	65.13	81.58	74.82	92.71	51.1	88.05	74.3
PAN	R-18	76.63	19.13	84.55	60.17	82.76	75.58	93.54	51.58	89.68	75.2
	R-34	78.17	14.27	86.35	60.22	84.46	80.42	93.23	55.21	90.22	75.27
	R-50	77.67	15.36	85.89	60.52	84.08	79.73	93.32	53.06	89.47	75.26
LinkNet	R-18	77.56	32.26	86.66	60.49	79.26	75.37	94.05	60.33	89.13	75.23
	R-34	79.09	21.46	88.14	62.49	86.3	82.3	94.59	58.94	87.74	72.23
	R-50	78.44	28.14	87.61	64.5	80.83	74.51	94.34	58.67	89.76	77.31
Unet+	R-18	78.63	23.27	87.56	64.94	81.11	77.94	94.43	59.3	89.7	74.02
	R-34	80.76	21.27	88.8	70.01	84.35	80.75	94.52	64.76	89.66	73.24
	R-50	79.61	28.85	89.24	65.64	83.47	77.54	94.06	63	87.96	75.93
Mean	/	77.43	23.15	86	62.28	82.26	76.53	93.59	55.9	88.73	74.16

In this study, we utilize Structural Similarity (SSIM), Peak Signal-to-Noise Ratio (PSNR), and the proposed Feature Entropy Index (FEI) to assess the effectiveness of the downsampling on the feature maps. Specifically, SSIM and PSNR are employed to evaluate the structural similarity and information fidelity between the input image and the feature maps. Additionally, the FEI and FEI boundary (FEI_B) are computed to quantify the information uncertainty in the downscaled feature maps and prediction outputs. In this study, a dilation morphological operation

with a kernel size of 5 is applied to extract object edges from the ground truth. The results in Table 4 demonstrate an improvement in SSIM (7.78%) and PSNR (2.14 dB) across all 21 models. Furthermore, the introduction of HWD in each architecture leads to a reduction in information uncertainty. Across all 21 models, the HWD reduces feature uncertainty by 58.2% (FEI) and 46.8% (FEI_B) comparing to the original downsampling methods. Furthermore, we observe that both the standard downsampling and HWD approaches exhibit higher uncertainty

Table 7

Performance of seven SOTA models with HWD on the Synapse test set.

Methods	Backbone	mDSC↑	mHD↓	Aorta	Gall	KidL	KidR	Liver	Pancreas	Spleen	Stomach
U-Net_HWD	R-18	79.52	21.22	87.8	63.44	84.97	79.62	93.91	60.58	89.76	76.12
	R-34	80.61	16.23	87.06	65.76	85.92	83.09	94.34	60.75	90.6	77.37
	R-50	78.6	26.95	87.32	64.14	83.02	79.99	94.06	58.69	88.52	73.08
DeepLabv3+_HWD	R-18	77.11	23.02	86.19	63.63	81.9	76.04	93.36	51.87	88.83	75.1
	R-34	77.82	22.09	87.3	61.56	82.61	78.69	93.69	53.39	89.74	75.57
	R-50	75.69	26.76	85.32	61.9	80.33	70.32	93.48	51.93	89.19	73.09
FPN_HWD	R-18	79.22	21.41	86.42	68.1	82.84	77.2	92.97	60.49	89.83	75.95
	R-34	79.38	16.6	86.16	69.67	82.83	76.92	92.89	60.38	89.59	76.63
	R-50	79.62	18.52	86.73	65.95	83.21	81.28	93.66	57.05	90.69	78.37
PSPNet_HWD	R-18	76.15	19.55	83.05	61.17	82.73	77.92	93.47	49.62	84.03	77.21
	R-34	76.94	18.3	82.51	65.29	82.83	77.46	93.49	51.25	87.13	75.52
	R-50	76.92	17.39	84.19	60.63	83.22	78.85	93.28	51.29	87.46	76.42
PAN_HWD	R-18	78.17	15.18	85.26	63.41	85.14	79.88	93.17	53.05	89.95	75.48
	R-34	77.78	20.17	85.88	64.41	81.77	75.77	93.01	55.02	90.15	76.22
	R-50	77.14	22.23	86.56	64.42	79.24	76.41	93	49.76	90.09	77.66
LinkNet_HWD	R-18	79.15	22.12	87.63	68.03	81.31	76.66	94.67	58.33	87.84	78.75
	R-34	79.59	24.1	87.72	62.23	83.19	79.97	94.29	63.23	89.07	76.97
	R-50	79.29	25.17	87.44	64.53	84.55	80.22	93.46	55.43	91.53	77.13
Unet+_HWD	R-18	80.24	22.73	87.52	72.52	83.75	78.95	94.11	61.39	89.57	74.14
	R-34	82.14	18.58	88.56	73.8	86.55	83.11	94.47	64.91	90.8	74.93
	R-50	80.85	24.07	88.35	67.31	84.02	79.61	94.04	65.05	91.52	76.89
Mean	/	78.66	21.07	86.43	65.33	83.14	78.47	93.66	56.83	89.33	76.12

Table 8

Comparison with the SOTA results w/o HWD on the MOST test set.

Model	Backbone	mDSC	Vessel	Soma	Model	mDSC	Vessel	Soma
U-Net	R-18	84.39	82.47	86.30	U-Net_HWD	87.40	81.54	93.25
	R-34	84.01	79.72	88.30		87.97	82.03	93.91
	R-50	88.30	83.27	93.33		88.38	83.60	93.15
DeepLabv3+	R-18	82.91	76.50	89.32	DeepLabv3+_HWD	86.32	80.72	91.92
	R-34	83.59	78.23	88.94		85.24	79.17	91.30
	R-50	84.00	77.57	90.43		86.29	80.46	92.11
FPN	R-18	86.86	82.39	91.32	FPN_HWD	86.75	82.60	90.89
	R-34	86.11	80.67	91.54		86.58	81.76	91.40
	R-50	86.77	82.04	91.49		86.82	81.94	91.69
PSPNet	R-18	83.86	78.96	88.76	PSPNet_HWD	83.86	79.03	88.68
	R-34	82.61	78.47	86.75		83.32	79.10	87.54
	R-50	83.46	78.35	88.56		82.12	75.88	88.35
PAN	R-18	87.96	83.75	92.17	PAN_HWD	88.53	83.74	93.32
	R-34	85.57	80.33	90.80		87.03	82.13	91.93
	R-50	86.37	80.29	92.45		88.65	85.00	92.30
LinkNet	R-18	85.67	77.57	93.76	LinkNet_HWD	86.48	79.02	93.94
	R-34	86.09	79.06	93.12		87.73	81.74	93.71
	R-50	85.36	77.23	93.49		89.10	84.32	93.87
Unet+	R-18	88.59	83.46	93.71	Unet+_HWD	89.17	84.87	93.47
	R-34	86.99	80.20	93.78		87.75	81.70	93.79
	R-50	88.39	82.96	93.81		88.85	83.77	93.92
Mean	/	85.61	80.17	91.05	Mean	86.87	81.62	92.12

near the object boundaries, indicating the difficulty in accurately segmenting these regions.

(3) Ablation study

We observed that state-of-the-art (SOTA) architectures equipped with HWD yield superior segmentation results compared to conventional methods such as max pooling or strided-convolution. Based on the decomposition ability of HWD, which separates a feature map into a low-frequency part (A) and three high-frequency components (D, V, and H) (Fig. 4), we conduct ablation studies to investigate the relative importance of each part in the semantic segmentation task.

Low frequency vs. high frequency: In Fig. 7, we assess the dominant component, after Haar wavelet transform, for achieving optimal segmentation performance. The low-frequency component (A) is found to be crucial for enhancing the segmentation performance of objects. This component significantly improves the mean Dice Similarity Coefficient (DSC) by 0.91% compared to using only high-frequency information (D, V, and H), despite the high-frequency components having three times the number of feature maps than their low-frequency counterpart.

The number of HWD modules in ResNet backbones:As demonstrated by the aforementioned experiments, replacing the downsampling module in the stem of ResNet with HWD significantly improves the segmentation performance by incorporating additional information into the models. The objective of this ablation study is to assess the effects of replacing the remaining downsampling layers in ResNets with HWD. In particular, ResNet architecture comprises of a stem and four stage blocks, consisting of convolution layers, batch normalization, and ReLU operations, along with skip connections. The stem involves two downsampling operations, namely, strided convolution and max pooling. In stage 2, stage 3, and stage 4, each includes two strided convolution operations-one in the skip connection and the other in the residual block. In this study, we replace the downsampling operations in various blocks of ResNet with HWD. Comparing with the original backbone, it is evident from Table 5 that utilizing HWD as the downsampling operation leads to improved segmentation performance. Furthermore, we observe that it is more effective to use HWD solely in the stem of ResNet. Therefore, we have implemented two HWD modules to replace the downsampling operations in the stem of ResNet, achieving

Table A.1

The definition of mathematical notations used in this paper.

Variable	Definition
$\tilde{a}$	The value after pooling
a_i	a_i represents the values in a region R of the input data
$ R $	$ R $ denotes the cardinality or the number of elements in the region R
ρ_i	The probability of a_i being selected undergoing stochastic pooling
H	The height of feature map
W	The width of feature map
H_0	The low-pass decomposition filters
H_1	The high-pass decomposition filters
ϕ	The wavelet basis function
ψ	The scale function
g	The ground truth
p	The prediction results
g_i	The pixels in g
p_i	The pixels in p
N	The number of image pixels
C	The number of label classes
TP	The number of positive instances that are correctly predicted
FP	The number of negative instances that are incorrectly predicted
TN	The number of positive instances that are incorrectly predicted
x	The point in set X
y	The point in set Y

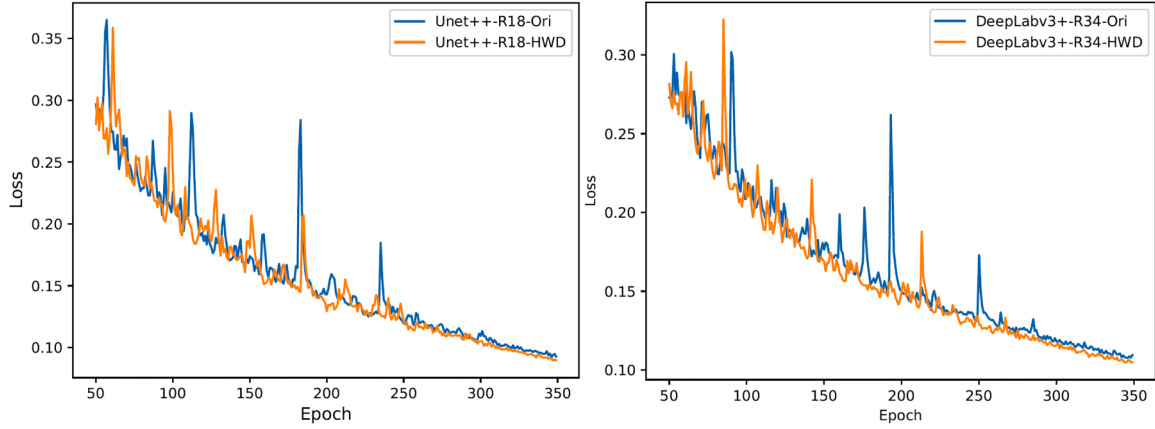
a favorable balance between efficiency and accuracy in this study.

4.4. Experiment results on Synapse dataset

We perform experiments using the same seven SOTA models as those employed in the Synapse dataset. The results demonstrate that integrating our HWD module into the architecture improves the DSC by 1.23% and reduces the HD by 2.08mm across the 21 models (Tables 6 and 7). Specifically, when compared to the seven models using R-50 as the backbone, our HWD module enhances the mean IoU by 1.57% when R-18 and R-34 are used as backbones. However, it only increases the mean IoU by 0.93% when R-18 and R-34 are used as backbones, indicating that our proposed method is more effective for architectures with fewer parameters.

4.5. Experiment results on MOST dataset

The segmentation architecture integrated with our proposed HWD module demonstrates a 1.26% improvement in terms of DSC (Table 8). Importantly, the HWD module significantly enhances the segmentation performance when ResNet-18 and ResNet-34 are used as backbones for feature extraction. For instance, the DSC values for U-Net_R18 and U-Net_R34 are 84.39% and 84.01%, respectively. However, upon integration of our HWD module, the DSC improves by 3.01% and 3.96% for U-Net_R18 and U-Net_R34, respectively. This demonstrates that shallow CNNs have a higher demand for information compared to relatively

**Fig. B.1.** Comparison of training loss with/without HWD in DeepLabv3+ architecture with ResNet34 as backbone.**Table C.1**

Mean performance of 21 segmentation models with ResNet as backbone, by using Max pooling, average pooling and HWD as downsampling operation on Camvid dataset.

Method	mIoU↑	Sky	BLD	Pol	Roa	Pav	Tre	Sig	Fen	Car	Ped	Bik
MaxPooling in HWD	60.77	90.32	78.92	23.33	94.31	81.05	69.66	33.31	25.31	82.15	45.91	44.19
AveragePooling in HWD	61.84	91.40	79.12	26.13	94.47	81.42	71.62	34.45	24.45	82.91	47.82	46.46
HWD	63.58	91.59	80.32	28.12	94.71	82.24	72.94	36.76	30.69	83.86	49.24	48.91

Table D.1

Performance of seven SOTA segmentation architectures as MobileNetv2 backbone on Camvid dataset.

Method	mIoU↑	Sky	BLD	Pol	Roa	Pav	Tre	Sig	Fen	Car	Ped	Bik
PSP	53.04	89.70	73.75	16.00	90.90	73.39	67.94	29.50	16.48	73.17	30.29	22.39
LinkNet	53.70	90.92	74.47	2.87	93.07	77.84	66.90	18.89	19.24	79.90	29.31	37.34
FPN	57.76	89.86	75.85	23.28	94.09	80.10	68.19	30.09	17.35	78.97	40.25	37.37
PAN	55.54	90.53	75.24	19.46	93.29	77.13	67.30	24.47	17.24	78.92	35.76	31.61
DeepLabv3+	56.07	91.04	75.26	20.16	93.42	78.63	68.15	26.04	18.18	74.66	34.89	36.37
U-Net	57.43	90.13	75.46	22.29	93.46	79.85	68.48	27.89	17.32	74.13	43.58	39.17
Unet+	59.00	90.49	76.93	25.09	93.46	79.49	69.60	30.38	21.30	79.30	42.07	40.89
Mean	56.08	90.38	75.28	18.45	93.10	78.06	68.08	26.75	18.16	77.01	36.59	35.02

Table D.2

Performance of seven SOTA segmentation architectures as MobileNetV2 backbone equipped with HWD module on Camvid dataset.

Method	mIoU↑	Sky	BLD	Pol	Roa	Pav	Tre	Sig	Fen	Car	Ped	Bik
PSP	53.81	89.60	74.25	16.21	91.41	73.69	69.85	27.23	24.90	72.71	31.65	20.38
LinkNet	55.72	89.97	75.15	8.26	93.13	77.62	68.06	22.09	19.61	82.93	34.81	41.34
FPN	58.38	91.07	77.54	22.59	93.16	77.58	69.51	28.85	19.76	80.03	40.53	41.57
PAN	55.76	90.60	73.94	19.50	93.59	78.46	66.94	26.44	15.09	76.54	32.93	39.28
DeepLabv3+	56.44	90.89	75.45	19.76	93.09	77.92	66.05	29.87	14.85	77.30	37.39	38.30
U-Net	58.89	91.71	77.01	24.03	93.48	79.94	68.10	30.59	24.83	76.16	41.59	40.38
Unet+	58.30	90.81	76.52	26.18	93.85	80.95	68.24	29.38	17.59	79.59	41.26	36.93
Mean	56.76	90.66	75.69	19.50	93.10	78.02	68.11	27.78	19.52	77.89	37.17	36.88

Table E.1

Performance of seven SOTA models without HWD on the Synapse test set.

Backbone	mDSC↑	mHD↓	Aorta	Gall	KidL	KidR	Liver	Pancreas	Spleen	Stomach
HRNet	77.10	21.36	87.61	62.69	79.71	75.65	93.49	54.20	89.79	73.65
HRNet_HWD	78.86	18.51	88.01	65.28	83.05	78.81	93.76	55.18	89.47	77.32
ConvNext	57.39	81.03	70.27	37.65	60.01	63.81	85.94	18.95	70.09	52.37
ConvNext_HWD	65.11	27.63	70.22	44.74	73.30	72.07	89.97	30.53	80.70	59.37
UNet	70.83	42.45	82.10	60.04	72.56	64.31	91.64	43.39	82.83	69.79
UNet_HWD	72.94	32.45	84.23	55.90	79.95	68.83	92.81	45.43	86.00	70.33
TransUNet	75.21	41.82	87.59	55.03	82.00	73.39	92.52	55.13	86.01	70.04
TransUNet_HWD	75.64	45.36	87.78	62.22	82.07	74.23	91.64	54.75	83.59	68.86
Swin-Unet	59.87	55.58	63.29	51.95	57.98	60.29	89.16	30.37	71.39	54.54
Swin-Unet_HWD	69.77	28.26	73.00	59.26	77.27	71.84	90.58	39.60	83.11	63.49
SegFormer	74.82	22.79	81.54	63.62	80.90	78.28	92.05	49.80	83.24	69.16
SegFormer_HWD	77.33	24.64	84.44	59.33	86.88	81.42	91.49	52.20	85.81	77.09
Mean Improved↑	4.07	14.69	2.55	2.63	8.23	5.24	0.91	4.31	4.22	4.49

deep networks.

5. Conclusion

In conclusion, we present a general downsampling module (HWD) for semantic segmentation in the paper. The goal of the HWD module is to retain as much essential information as possible during downsampling. Extensive experiments and ablation studies conducted on three different image datasets with varying modalities demonstrate the effectiveness of the proposed HWD module and the FEI metric. This work has implications for various CNN-based computer vision tasks, including instance segmentation, object detection, and pose estimation. Furthermore, to assess the quality of downsampled feature maps, we introduce a new metric called Feature Entropy Index (FEI). The FEI metric effectively reflects the degree of information uncertainty by considering the downsampled feature maps and the prediction results. Experimental results further indicate that the HWD module provides more information for object segmentation compared to conventional downsampling methods.

The proposed HWD downsampling module and FEI assessment metric have the following two main advantages. Firstly, the HWD module can seamlessly integrate into existing segmentation architectures due to its generality. It can directly replace existing downsampling methods, such as max pooling, average pooling, or strided convolution, without introducing additional complexities. Furthermore, it significantly improves the segmentation performance. Secondly, regarding the FEI, it can be applied to assess the quality of feature maps, serving as a quantitative indicator for assessing the amount of essential information preserved after downsampling in segmentation architectures.

The primary limitations of the proposed HWD module lie in its efficiency and locality. Therefore, we intend to continue our research in two aspects. First, it is crucial to extract representative features from input images for semantic segmentation. However, the source images contain a significant amount of redundant information, which may hinder the extraction of representative features. We aim to incorporate prior knowledge, such as boundaries and textures, into the HWD module to efficiently filter out irrelevant information during the downsampling

process. Second, the proposed HWD offers a great potential for improving segmentation performance on the current SOTA convolutional neural network benchmarks, like U-Net, DeepLabv3+ and PSPNet. However, the HWD module lacks the ability to capture global context and establish long-range spatial relations due to the localized nature of convolution operations. We are considering integrating both local and global features from convolution and Transformer operations into the HWD module in our future work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work is supported by the Guangdong Provincial Key Laboratory of Human Digital Twin (No. 2022B1212010004), the Open-Fund of WNLO (No. 2018WNLOKF027), the Hubei Key Laboratory of Intelligent Robot in Wuhan Institute of Technology (No. HBIRL202202 and No. HBIR202206), and the Chongqing Science and Technology Bureau (No. 2022TIAD-KPX0190). We thank the Optical Bioimaging Core Facility of WNLO-HUST for the support in MOST data acquisition.

Appendix A. mathematical notations table

Appendix B. Training Loss with/without HWD

This section presents the training loss with and without HWD in the ResNet backbones. As shown in Fig. B.1, the training loss exhibits a

slightly faster decrease when utilizing the HWD module in the Unet++ and DeepLabv3+ architectures compared to the original ResNet backbone. Specifically, the training loss is consistently lower when employing the HWD module compared to using conventional downsampling operations such as strided convolution and max pooling.

Appendix C. Replace lossless feature encoding block of HWD with max pooling or average pooling

We also conduct experiments by replacing the lossless feature encoding block (haar wavelet transform) of HWD with max pooling layer or average pooling, in order to evaluate the performance improvement of HWD not only by increasing the representation feature learning block of HWD. Table C.1 shows the performance of mean IoU of total 21 segmentation models. These models comprise seven semantic segmentation architectures that employ three different types of ResNet as their backbones. It can be observed that the Haar wavelet transform plays a more significant role compared to max pooling and average pooling, resulting in improvements of 2.81% and 1.74% in mean IoU across all 21 segmentation models.

Appendix D. MobileNetv2 as backbone under seven segmentation architectures

This section contains detailed results for CamVid dataset with Mobilenetv2 [65] as backbone under seven semantic segmentation frameworks. MobileNetv2 was originally developed for mobile devices, striking a balance between accuracy, latency, and parameter count for classification tasks. There are 5 times downsampling operations in the MobileNetv2. The first downsampling (strided convolution) happens in the stem of MobileNetv2, and the others are in the bottleneck of it. Here, we only use HWD to replace the first strided convolution operation by the considering of the accuracy and efficiency.

Tables D.1 and D.2 show the segmentation results on seven segmentation architectures with and without HWD in MobileNetv2 on CamVid testset in terms of mean IoU. We can find that MobileNetv2 equipped with HWD could achieve the performance improvement of 0.8% mIoU on seven segmentation architectures, demonstrating our method could be adapted in other backbones too. Specifically, our method could improve the performance of small-scale objects in a large margin, such as pole (1.05%), sign symbol (1.03%), fence (1.36%) and bicycle (1.86%).

Appendix E. Performance with the SOTA architectures based convolution and transformer on synapse dataset

In this section, we test six SOTA segmentation methods on Synapse dataset. They are HRNet [66], ConvNeXt [67], U-Net [68], TransUNet [60], Swin-Unet [69], and SegFormer [70], respectively. Note that the HRNet and ConvNeXt are pure convolution neural network, the U-Net and TransUNet are hybrid with convolution and transformer operations. Besides, the Swin-Unet and SegFormer are based on transformer. We tested these methods on Synapse dataset, and the comparative results are shown in Table E.1. Here, we simply replace the downsampling operation, like maxpooling or convolution stride, with our proposed HWD to reduce the resolution of feature maps on each architecture.

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